
Machine Learning Based Analog Circuit Fault Diagnosis
PDF 7 — Chapter 13: GridSearchCV
Hyperparameter Tuning | Cross Validation | Implementation Line-by-Line

PART 1 — Why We Needed Hyperparameter Tuning

Chapter 13 — GridSearchCV Part 1 — Why We Needed Hyperparameter Tuning Before Starting Congratulations. Until now, our pipeline looks like this. **LTspice** — CSV Files.

17 Features

StandardScaler — SVM. Question. Is our SVM ready? Not yet. One very important question still remains. Which SVM should we use? Most beginners think there is only one SVM. Wrong. An SVM has many settings. Question. Who chooses those settings? That is exactly what GridSearchCV does.

What you will know after finishing this chapter

After finishing this chapter, you should be able to answer [OK] What are hyperparameters? [OK] Why can't we manually choose them? [OK] What is GridSearchCV? [OK] Why did we use it? [OK] Why is it better than trial-and-error?

Difficulty

Basics 5/5

Theory 5/5

Coding 3/5

Viva Importance 5/5

Let's Begin

Suppose I give you two identical cars. Question. Will they perform the same? Not necessarily. Why? Because one may have different Tyres

Engine Tuning

Suspension

Gear Ratio

Same car. Different settings. Exactly the same happens with SVM. Does SVM Have Settings? Yes. Question. Which settings? The important ones are Kernel C Gamma Question. Are these learned during training? No. These must be chosen before training starts. These are called Hyperparameters.

PROFESSOR ASKS

What is a Hyperparameter?

EXCELLENT ANSWER

A hyperparameter is a configuration value that is set before training and controls how the machine learning algorithm learns from the data. Notice the difference. **Parameters** — Learned. **Hyperparameters** — Chosen before learning. Very important. Parameter vs Hyperparameter Students always confuse these. Let's compare.

Learned during training	Set before training
Example: Support Vectors	Example: C, Gamma
Model discovers them	User chooses them

Remember this table. Very important. Why Not Choose Randomly? Question. Suppose I choose

C = 1000

Gamma = 1000

Question. Will the model be good? Maybe. Maybe not. Question. Suppose I choose

C = 0.0001

Gamma = 0.0001

Again Maybe. Maybe not. Question. How do we know which values are best? Need systematic testing. Trial-and-Error One method is trial-and-error. Example. Try

C = 1

Train. Check accuracy. Now Try

C = 10

Train again. Then Try

C = 100

Again. Question. Is this reliable? No. Because we might miss better values.

Real Life Analogy

Imagine buying shoes. Question. Would you try only one size? No. You try many sizes until one fits perfectly. GridSearchCV does exactly the same thing. What is Grid Search? Question. Suppose we have **C** — 1. 10 100 and **Gamma** — 0.1. 0.01 0.001 Question. How many combinations exist? Let's see. C=1 Gamma=0.1 C=1 Gamma=0.01 C=1 Gamma=0.001 C=10 Gamma=0.1. Every possible combination is tested. That is called Grid Search.

PROFESSOR ASKS

Why is it called Grid Search?

EXCELLENT ANSWER

Grid Search systematically evaluates every possible combination of predefined hyperparameter values to identify the best-performing configuration. Excellent. Why "CV"? Question. What does CV mean? Answer. Cross Validation. Question. What is Cross Validation? We'll study it in detail in the next part. For now, remember

Grid Search

+

Cross Validation

= GridSearchCV Why Didn't We Choose Parameters Manually? Question. Could we have manually selected C and Gamma? Yes. Question. Would it be scientifically sound? No. Because our personal choice may not be optimal. Instead we allowed GridSearchCV to evaluate all combinations objectively. That makes our project more rigorous.

PROFESSOR ASKS

Why did you use GridSearchCV?

EXCELLENT ANSWER

We used GridSearchCV to systematically evaluate multiple hyperparameter combinations and automatically select the configuration that produced the best cross-validation performance. This is an excellent research answer.

In Our Project

Instead of saying I think C=10 looks good. We said Let GridSearchCV evaluate all candidate values and choose the best one. That is good engineering.

Real Engineering Thinking

Notice something. Throughout our project we never guessed. **Dataset** — Generated systematically. **Features** — Engineered systematically. **DC Level** — Added. after error analysis. **SVM Parameters** — Selected. using GridSearchCV. Everything was data-driven. This is one of the biggest strengths of your project.

Connection to Our Project

Our pipeline now becomes

17 Features

StandardScaler — GridSearchCV. **Best SVM** — Training. Prediction Notice. GridSearchCV comes before the final model is trained.

COMMON MISTAKES

MISTAKE: [X] GridSearchCV trains only one model. Wrong. It trains many models.

MISTAKE: [X] Hyperparameters are learned during training. Wrong. They are selected before training.

MISTAKE: [X] Choosing random values is enough. Wrong. Systematic optimization is much better.

REVISION NOTES

Remember these ideas.

Hyperparameter — Algorithm Setting.

Grid Search — Tests Every Combination.

CV — Cross Validation.

GridSearchCV — Automatic Hyperparameter Selection.

MEMORY TRICK

Remember this sentence. "Grid Search tries everything. Cross Validation checks everything. Together they choose the best model." One sentence. Entire chapter.

KNOWLEDGE CHECK

Level 1

- What is a hyperparameter?
- What is Grid Search?
- What does CV stand for?
- Why did we use GridSearchCV?
- Why didn't we manually choose C and Gamma?

Level 2

- Explain the difference between parameters and hyperparameters.
- Why is GridSearchCV more reliable than trial-and-error?
- How does Grid Search explore the parameter space?
- Why is systematic optimization important in research?
- At what stage of our project is GridSearchCV used?

Professor Level

- Explain the complete purpose of GridSearchCV in your project.
- Why is hyperparameter optimization necessary for SVM?
- What are the limitations of manual hyperparameter selection?
- How does GridSearchCV contribute to the reliability of experimental results?
- Why is hyperparameter tuning considered part of the machine learning pipeline rather than model evaluation?
- End of Chapter 13 (Part 1)
- Congratulations.
- You now understand why GridSearchCV exists and why it was an essential part of your project. More importantly, you can now explain that your SVM configuration was not chosen arbitrarily—it was selected through a systematic hyperparameter optimization process.

Author's Note The next part is where GridSearchCV becomes even more interesting. We'll study: Cross Validation in depth. Why using a single train-test split can sometimes give misleading results. How K-Fold Cross Validation works. How GridSearchCV combines Grid Search with Cross Validation. The complete workflow used in your project to identify the optimal SVM configuration. This is one of the topics that often impresses interviewers because it demonstrates that your model selection process followed sound machine learning practice rather than relying on a single lucky split of the data.

PART 2 — Cross Validation

Chapter 13 — GridSearchCV Part 2 — Cross Validation (The Secret Behind Reliable Machine Learning) Before Starting

In Part 1

we learned that

Grid Search

tries many combinations of Kernel C Gamma Question. Is that enough? No. Suppose one parameter gets 98% accuracy. Question. Can we immediately say it is the best? Not necessarily. Why? Because maybe our train-test split was simply lucky. Need something more reliable. That is Cross Validation.

First Question

Suppose we have 1000 samples. Question. What did we do earlier? Simple. 80% **Training** — 20%. Testing Question. Is this always reliable? Imagine the 20% testing set happens to contain only easy samples. Accuracy becomes very high. Now imagine another split contains many difficult samples. Accuracy falls. Question. Did the model change? No. Only the split changed. Very important.

PROFESSOR ASKS

What is the limitation of a single train-test split?

EXCELLENT ANSWER

A single train-test split may produce misleading results because the evaluation depends on one particular division of the dataset, which may not represent the overall data distribution. Excellent.

Real Life Analogy

Imagine a cricket player. Question. Can we judge his career based on one match? No. Maybe he had a bad day. Need many matches. Then take the average. Exactly what

Cross Validation

does.

The Main Idea

Instead of testing only once, we test many times. Each time using a different training and testing split. Finally we average all results. Now our evaluation becomes much more reliable.

K-Fold Cross Validation

The most common method is K-Fold Cross Validation. Question. What is K? K means the number of folds. Suppose K = 5. Question. What happens? Our dataset is divided into five equal parts. Example

Fold 1

Fold 2

Fold 3

Fold 4

Fold 5 Round 1 Test

Fold 1

Train

Fold 2

Fold 3

Fold 4

Fold 5

Train. Evaluate. Store accuracy.

Round 2

Now change the testing fold. Train

Fold 1

Test

Fold 2

Train

Fold 3

Fold 4

Fold 5

Again train. Again evaluate. Round 3 Train

Fold 1

Fold 2

Test

Fold 3

Train

Fold 4

Fold 5

Again. Eventually every fold becomes the testing set exactly once.

Final Step

Suppose accuracies are 96% 97% 95% 98% 96% Question. Which accuracy do we report? Not 98%. Not 95%. We report the average. Average = 96.4% This is much more reliable.

PROFESSOR ASKS

What is K-Fold Cross Validation?

EXCELLENT ANSWER

K-Fold Cross Validation divides the dataset into K equal folds. Each fold is used once for testing while the remaining folds are used for training. The final performance is obtained by averaging the results across all folds. Excellent. Why Is This Better? Question. Suppose one fold contains many difficult samples. Question. Will it ruin our evaluation? No. Because the other folds balance it. This makes our estimate much more stable.

Real Life Analogy

Imagine five professors checking your project. Question. Would you trust the opinion of one professor or the average of all five? Obviously the average.

Cross Validation

works exactly the same way.

Now Combine Everything

Question. What does Grid Search do? It tests many values of Kernel C Gamma Question. What does

Cross Validation

do? It checks each combination using multiple train-test splits. Together they become GridSearchCV. Inside GridSearchCV Suppose we have **Kernel** — RBF. Polynomial and **C** — 1. 10 Question. How many

combinations? Four. Now suppose we use 5-fold CV. Question. How many trainings? Each combination is trained five times. Total $4 \times 5 = 20$ trainings Notice why GridSearchCV takes time. It trains many models.

In Our Project

This is exactly what happened. Our code did not simply say

C = 10

Instead GridSearchCV evaluated multiple parameter combinations, performed

Cross Validation,

calculated the average score, and selected the best model automatically. That model was then used for our final fault classifier.

PROFESSOR ASKS

How did GridSearchCV choose the best model?

EXCELLENT ANSWER

GridSearchCV evaluated multiple hyperparameter combinations using cross-validation, calculated the average validation performance for each combination, and selected the configuration with the highest average score. Excellent. Why Is This Scientifically Better? Imagine writing a research paper. Question. Which statement sounds stronger?

Statement 1

I tried $C = 10$. It worked.

Statement 2

Multiple parameter combinations were evaluated using GridSearchCV with cross-validation, and the best-performing configuration was selected automatically. Question. Which sounds more scientific? Obviously Statement 2. That is exactly what your project did. Where Does Cross Validation End? Very important. Question. After GridSearchCV selects the best parameters, what happens? Now we train the final SVM using those best parameters on the training data. Only then do we evaluate it on the untouched test set. Notice the test set is still kept separate. This prevents data leakage.

PROFESSOR ASKS

Why should the test set remain untouched during GridSearchCV?

EXCELLENT ANSWER

The test set must remain unseen during hyperparameter tuning to ensure an unbiased evaluation of the final model's generalization performance. This is a very strong viva answer.

Connection to Our Project

Our pipeline now becomes

17 Features

Train-Test Split

GridSearchCV

Cross Validation

Best Parameters

Final SVM Training — Prediction. Evaluation This is exactly the sequence followed in your project.

COMMON MISTAKES

MISTAKE: [X] Cross Validation replaces the test set. Wrong. The final test set is still required.

MISTAKE: [X] GridSearchCV tests only one model. Wrong. It trains many models using many folds.

MISTAKE: [X] Highest accuracy from one fold is reported. Wrong. The average across folds is used.

REVISION NOTES

Remember these ideas.

Single Split — May be misleading.

K-Fold — Multiple evaluations.

Average Score — Reliable estimate.

GridSearchCV — Grid Search

Cross Validation.

Test Set — Untouched until final evaluation.

MEMORY TRICK

Remember this sentence. "Grid Search finds candidates. Cross Validation tests them fairly. The best average wins." One sentence. Entire chapter.

KNOWLEDGE CHECK

Level 1

- What is Cross Validation?
- What is K-Fold Cross Validation?
- Why is a single train-test split sometimes unreliable?
- What does GridSearchCV combine?
- Why do we average the fold accuracies?

Level 2

- Explain the complete K-Fold Cross Validation process.
- Why should the test set remain untouched during hyperparameter tuning?
- How does GridSearchCV avoid biased parameter selection?
- Why is the average validation score preferred over a single fold score?
- Why does GridSearchCV take longer than training one SVM?

Professor Level

- Explain the complete hyperparameter optimization pipeline used in your project.
- Why is Cross Validation considered a more reliable estimator of model performance?
- What is the difference between the validation folds and the final test set?
- How would the results change if the test set were accidentally included during GridSearchCV?
- Why is GridSearchCV widely used in machine learning research?
- End of Chapter 13 (Part 2)
- Congratulations.
- You now understand how GridSearchCV and Cross Validation work together to produce a reliable and scientifically sound SVM configuration.
- More importantly, you can explain why your project did not rely on guesswork. Every important SVM hyperparameter was selected through a systematic evaluation process, which is a strong point in both a viva and a technical interview.

Author's Note The next part is where we'll connect this theory directly to your code. We'll cover: The actual GridSearchCV(.) call used in your implementation. Every argument (param_grid, cv, scoring, n_jobs, etc.). How the best parameters were extracted. How the optimized SVM model was created. Why this pipeline made your analog fault diagnosis system more robust and reproducible. That chapter will bridge the gap between the concepts you've learned and the exact implementation in your project.

PART 3 — Our GridSearchCV Implementation

Chapter 13 — GridSearchCV Part 3 — Our GridSearchCV Implementation (Line-by-Line Explanation)
Before Starting In the previous chapter, we understood the theory behind GridSearchCV. Now we'll study what our code actually does. This chapter is extremely important because many students can explain GridSearchCV, but very few can explain their own implementation. After this chapter, you'll be able to explain every line of your tuning pipeline. Where Are We in the Pipeline? Let's see our project again. **LTspice** — CSV Files.

17 Features

Train-Test Split

StandardScaler — GridSearchCV ← We are here. **Best SVM** — Prediction. Everything before this has already happened. Now we need to find the best SVM.

Step 1 — Creating the Parameter Grid

Question. How does GridSearchCV know which values to test? Simple. We give it a list. Conceptually, our code looks like `param_grid = { "C": [1], "gamma": [1], "kernel": [1] }` Question. What is `param_grid`? It is simply a dictionary containing all candidate values that GridSearchCV will test.

PROFESSOR ASKS

What is `param_grid`?

EXCELLENT ANSWER

The parameter grid is a collection of candidate hyperparameter values that GridSearchCV systematically evaluates.

Real Life Analogy

Imagine ordering a pizza. Question. What choices do you have? **Size** — Small. Medium Large **Crust** — Thin. Cheese Burst **Toppings** — Veg. Paneer Corn Question. How many possible pizzas exist? Many. GridSearchCV does exactly the same thing. It creates every possible combination.

Step 2 — Creating GridSearchCV

Now our code creates the optimizer. Conceptually, it looks like

```
grid = GridSearchCV( estimator=model, param_grid=param_grid, cv=5
)
```

Let's understand every parameter. estimator Question. What is the estimator? Answer. The estimator is the model we want to optimize. In our project, the estimator is our SVM.

PROFESSOR ASKS

What is estimator?

EXCELLENT ANSWER

The estimator is the machine learning model whose hyperparameters are being optimized. `param_grid` Already studied. Contains all candidate values. `cv` Question. What does `cv=5` mean? Simple. Five-fold Cross Validation. Exactly what we studied in the previous chapter. Question. Does GridSearchCV train only once? No. Suppose there are 20 parameter combinations. Question. How many trainings? $20 \times 5 = 100$ trainings Notice why GridSearchCV takes time. scoring Many students ignore this. Question. Suppose we have Accuracy Precision Recall

F1 Score

Question. Which metric should GridSearchCV maximize? Answer. The scoring parameter tells it. In our project, accuracy was used for model selection. (We'll verify the exact metric from your code when we reach the code appendix.)

PROFESSOR ASKS

Why scoring?

EXCELLENT ANSWER

The scoring parameter specifies the evaluation metric used by GridSearchCV to compare different hyperparameter combinations. `n_jobs` Another useful parameter. Question. Suppose your computer has 8 CPU cores. Question. Should GridSearchCV use only one? No. Using multiple cores makes training much faster. That's what `n_jobs` controls. `verbose` Question. Suppose GridSearchCV is running. Question. Would you like to see its progress? Yes. Verbose prints progress messages. Helpful during long training.

Step 3 — Running GridSearchCV

Now the most important line. Conceptually, our code executes `grid.fit(X_train, y_train)` Question. Is this the same as `model.fit()`? Not exactly. Question. Why? Because GridSearchCV internally calls `model.fit()` many, many times. Remember every combination $\times$ every fold.

PROFESSOR ASKS

What happens during `grid.fit()`?

EXCELLENT ANSWER

GridSearchCV trains multiple SVM models using different hyperparameter combinations and cross-validation folds before selecting the best-performing configuration. Excellent.

Step 4 — Finding the Best Parameters

Question. After GridSearchCV finishes, how do we know the winner? Simple. Conceptually, our code uses `grid.best_params_` Question. What does `best_params_` contain? Something like **Kernel** — RBF. **C** — 10. **Gamma** — 0.01. (Not necessarily these exact values.) These are the best hyperparameters found during Grid Search.

PROFESSOR ASKS

What is `best_params_`?

EXCELLENT ANSWER

`best_params_` stores the hyperparameter combination that achieved the highest average cross-validation score.

Step 5 — Best Score

Question. How good was the best model? GridSearchCV stores that too. Conceptually, `grid.best_score_` This is the average

Cross Validation

performance of the winning model. Remember this is NOT the final test accuracy. Very important.

Difference Between

`best_score_` and

Test Accuracy

Students always confuse these. Let's compare.

Cross Validation average	Final unseen test performance
Used during tuning	Used after tuning
Chooses best model	Evaluates final model

Very important.

Step 6 — Creating the Final Model

Question. Suppose GridSearchCV finds the best parameters. Question. Do we continue using GridSearchCV for prediction? Usually No. Instead, we take the best estimator. Conceptually, `best_model = grid.best_estimator_` Now this becomes our final classifier.

PROFESSOR ASKS
What is <code>best_estimator_</code> ?
EXCELLENT ANSWER

`best_estimator_` is the optimized machine learning model obtained after GridSearchCV finishes hyperparameter tuning.

Step 7 — Final Prediction

Now everything becomes normal. Our code simply executes `best_model.predict(X_test)` Notice GridSearchCV is finished. Hyperparameter tuning is complete. Now the optimized SVM predicts fault classes.

Complete Flow

Everything we learned can be summarized using one diagram.

Create Parameter Grid

Create GridSearchCV — `grid.fit()`.

Cross Validation

Best Parameters

Best Estimator

Prediction This is exactly the sequence followed inside our project.

Why This Makes Our Project Strong

Many student projects simply write

```
model = SVC(C=10)
```

Question. Why `C=10`? Answer. "I guessed." Weak answer. Our project says **GridSearchCV** — Evaluated. **all combinations** — Selected. the best one. This is research-quality engineering.

Connection to Our Project

Our complete Machine Learning pipeline now looks like this. **LTspice** — CSV Files.

17 Features

Train-Test Split

StandardScaler — GridSearchCV. **Best SVM** — Prediction. Evaluation Everything fits together.

COMMON MISTAKES

MISTAKE: [X] `best_score_` is the final accuracy. Wrong. It is the average Cross Validation score.

MISTAKE: [X] `GridSearchCV` replaces the final model. Wrong. It helps find the final model.

MISTAKE: [X] `best_params_` contains predictions. Wrong. It contains hyperparameter values.

REVISION NOTES

Remember these steps.

Parameter Grid — Candidate Values.

GridSearchCV — Optimizer.

grid.fit() — Many Trainings.

best_params_ — Best Hyperparameters.

best_estimator_ — Final Optimized Model.

MEMORY TRICK

Remember this sentence. "Grid builds candidates, fit tests them, best_params chooses them, best_estimator uses them." One sentence. Entire implementation.

KNOWLEDGE CHECK

Level 1

- What is `param_grid`?
- What is `estimator`?
- What does `grid.fit()` do?
- What is `best_params_`?
- What is `best_estimator_`?

Level 2

- Why is `best_score_` different from test accuracy?
- Why does `GridSearchCV` take longer than normal training?
- What happens after `GridSearchCV` finds the best parameters?
- Why is `best_estimator_` used for prediction?
- Why is `GridSearchCV` considered a systematic optimization technique?

Professor Level

- Explain the complete GridSearchCV implementation used in your project.
- Why is param_grid necessary?
- Why is Cross Validation performed inside GridSearchCV?
- What is the difference between model.fit() and grid.fit()?
- How does GridSearchCV improve the reproducibility of machine learning experiments?
- End of Chapter 13
- Congratulations.
- You now completely understand GridSearchCV, from theory to implementation.
- You can confidently explain:
 - Why hyperparameter tuning is necessary.
 - How Cross Validation works.
 - How GridSearchCV evaluates candidate models.
 - How the best SVM is selected.
 - And how your project systematically optimized its classifier.
- Author's Note
- The next chapter is one of my favorites because it's where machine learning becomes engineering.
- Chapter 14 — Model Evaluation
- We'll cover:
 - Confusion Matrix (in depth)
 - Accuracy
 - Precision
 - Recall
 - F1 Score
 - ROC Curve
 - AUC
 - Classification Report
 - Why accuracy alone is not enough
 - How the RC_Open problem was discovered from the confusion matrix
 - How evaluation led directly to the addition of DC Level and later the Topology Rule
 - This chapter is the turning point of your project. Up to now, we've been building the model. From the next chapter onward, we'll analyze, criticize, and improve it—the way real machine learning engineers do.